

WALLA WALLA, WA, USA

WMO: 727846

Lat: 46.095N

Lon: 118.287W

Elev: 355

StdP: 97.13

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 24160

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-11.0	-7.7	-14.9	1.1	-9.9	-11.4	1.5	-6.1	14.3	12.2	12.5	11.2	2.2	180	0.549

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.8	37.0	19.0	34.9	18.2	32.8	17.6	20.2	33.7	19.1	32.3	18.3	30.8	3.6	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.6	11.6	23.2	14.2	10.5	22.7	13.1	9.8	22.4	59.0	33.8	55.6	32.5	52.8	30.7	30.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.9	9.0	8.0	DB	-12.6	40.6	5.3	2.0	-16.4	42.0	-19.5	43.1	-22.4	44.3	-26.3	45.7	(4)
(5)			WB	-13.1	22.3	5.0	2.1	-16.7	23.8	-19.7	25.1	-22.5	26.3	-26.2	27.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	12.6	2.6	4.3	8.3	11.4	15.8	19.6	25.0	24.2	19.2	12.1	5.9	1.9	(6)	
	DBStd	8.91	5.33	4.69	3.69	3.34	3.98	3.82	3.49	3.38	3.75	3.73	4.87	5.27	(7)	
	HDD10.0	909	234	164	76	20	2	0	0	0	1	20	137	256	(8)	
	HDD18.3	2646	489	393	312	208	100	28	1	2	34	196	373	511	(9)	
	CDD10.0	1848	3	4	22	63	182	288	466	441	276	85	14	4	(10)	
	CDD18.3	542	0	0	0	1	22	66	208	184	59	2	0	0	(11)	
	CDH23.3	6310	0	0	2	23	253	759	2562	2094	594	23	1	0	(12)	
	CDH26.7	3034	0	0	0	2	78	308	1379	1048	216	3	0	0	(13)	
(14) Wind	WSAvg	3.6	3.2	3.5	4.0	4.0	3.8	3.9	3.8	3.7	3.4	3.3	3.4	3.3	(14)	
(15) Precipitation	PrecAvg	501	56	45	50	43	46	28	18	21	23	42	67	61	(15)	
	PrecMax	720	176	103	118	98	168	85	120	83	59	104	125	129	(16)	
	PrecMin	328	16	5	10	6	7	6	0	0	0	0	9	9	(17)	
	PrecStd	95	34	24	26	22	35	22	23	22	19	28	31	31	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	16.1	17.6	21.7	26.1	31.8	36.3	40.0	38.7	34.1	26.3	19.9	17.2	(19)	
		MCWB	9.9	9.7	12.2	15.6	18.0	19.7	19.7	18.9	18.0	15.1	13.0	10.5	(20)	
	2%	DB	13.5	14.3	18.5	23.0	29.0	32.7	37.4	36.4	31.4	22.7	16.9	13.9	(21)	
		MCWB	8.4	9.3	10.9	13.9	17.1	18.2	19.3	18.6	16.9	13.5	10.6	9.0	(22)	
	5%	DB	11.7	12.4	16.4	20.5	26.2	30.2	35.7	34.2	29.1	20.2	14.7	11.6	(23)	
		MCWB	7.8	8.2	9.9	12.4	15.5	17.1	18.4	17.7	16.2	12.6	10.0	7.8	(24)	
	10%	DB	9.8	10.7	14.4	17.9	23.5	27.7	33.7	32.3	27.0	18.0	12.6	9.3	(25)	
		MCWB	6.6	7.0	8.9	11.0	14.4	16.1	17.8	17.2	15.2	11.6	8.7	6.3	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.4	12.2	13.4	16.6	19.3	21.1	22.8	20.5	19.5	16.3	14.5	11.1	(27)	
		MCDB	14.3	14.6	19.3	24.2	29.2	32.9	35.7	35.7	30.5	22.7	18.5	16.0	(28)	
	2%	WB	9.2	10.0	11.7	14.4	17.7	19.3	20.5	19.5	18.1	14.7	11.9	9.7	(29)	
		MCDB	12.7	13.7	16.8	21.9	27.3	30.8	34.6	34.0	29.2	20.8	15.7	13.4	(30)	
	5%	WB	7.9	8.4	10.6	12.9	16.4	17.9	19.3	18.6	16.9	13.4	10.4	7.9	(31)	
		MCDB	11.1	11.8	15.4	19.3	24.6	28.3	33.2	32.3	26.8	18.9	13.5	11.2	(32)	
	10%	WB	6.6	7.2	9.5	11.6	15.2	16.7	18.4	17.8	15.8	12.3	9.0	6.4	(33)	
		MCDB	9.6	10.4	14.2	17.2	22.2	26.1	31.7	30.7	25.2	17.3	12.1	9.2	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	5.8	7.1	9.4	10.7	11.7	12.8	14.8	14.2	12.8	10.1	7.0	5.7	(35)	
		MCDBR	8.9	9.2	12.0	14.4	15.6	16.7	17.3	16.8	16.0	12.9	9.5	8.7	(36)	
	5% WB	MCWBR	5.5	5.7	6.7	7.7	7.2	6.7	6.5	6.5	6.7	6.5	5.3	5.6	(37)	
		MCWBR	8.5	8.7	11.0	13.8	14.2	15.5	16.2	15.6	14.4	11.5	8.3	8.4	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.289	0.294	0.305	0.334	0.352	0.353	0.354	0.357	0.337	0.317	0.294	0.285	(40)	
		taud	2.528	2.537	2.537	2.440	2.412	2.434	2.441	2.423	2.462	2.523	2.541	2.513	(41)	
	Ebn at Noon		828	897	932	925	914	912	906	890	881	853	813	790	(42)	
		Edh at Noon	64	76	88	106	113	111	109	107	94	76	62	58	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.23	2.24	3.50	5.14	6.26	7.09	7.66	6.43	4.75	2.93	1.54	0.99	(44)	
	RadStd		0.14	0.28	0.28	0.34	0.44	0.38	0.27	0.36	0.29	0.23	0.13	0.12	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	-0.54	N/A	N/A	N/A	N/A	N/A
(47) Regional (18 neighbors)		N/A	N/A	N/A	N/A	-0.33	N/A	N/A	N/A	+90	N/A

Nomenclature: See separate page